Year 10 Chemistry Atomic Structure Revision

2.1 Atoms, elements and compounds

1. What are the two main parts of an atom?

   _____

2. How do we define an element?

   _____

   _____

3. How many elements have been found in the earth's crust? _____
4. About how many elements have been made by scientists? _____
5. How many letters may be used for the symbol of an element? _____ or _____
6. Why is the symbol for

   sodium Na? _____

   lead Pb? _____

   potassium K? _____
7. Elements may be separated into two main types,

   _____ and _____
8. About how many of the elements are metals? _____
9. List three properties of metals.

   _____

   _____

   _____

   _____

   _____

   _____
10. About how many of the elements are non-metals?

    _____
11. List three properties of non-metals.

    _____

    _____

    _____

    _____

    _____

    _____

12. What is a compound?

_____

_____

13. There are only about 118 elements, but there are millions of compounds.
Why are there so many compounds?

_____

_____

_____

14. An element is represented by a _____
A compound is represented by a _____

15. Write the formulae for
water _____ alcohol (ethanol) _____
carbon dioxide _____ carbon monoxide _____
copper sulfate _____ nitrogen dioxide _____
common salt (sodium chloride) _____

16. What elements (names and symbols) are contained in the following substances?
ethanol _____
water _____
sodium bicarbonate, $NaHCO_3$

_____

17. Many compounds consist of molecules, which are groups of atoms joined together. Sketch a diagram that shows how the atoms are connected in H_2SO_4.

Sulphuric acid
H_2SO_4
is the acid in car batteries

18. Many formulae contain small numbers, eg; H_2SO_4
What do these numbers tell us about the compound?

_____

_____

19. Consider the formula for ordinary sugar (sucrose), $C_{12}H_{22}O_{11}$.
What does this tell you about the compound, and about a molecule of sugar?

_____

_____

_____

2.2 Atoms, elements and compounds

Subatomic particles are the parts that make up an atom.

1. What two types of subatomic particles are in the nucleus?

   _____ and _____

2. What units are used to measure the masses of subatomic particles?

   _____ (amu)

3. Complete this table that describes the masses, charges and locations of protons, neutrons and electrons.

Subatomic particle	Mass (amu)	Charge (+1,–1 or 0)	Location (in, or around, nucleus)
Proton			
Neutron			
Electron	1 / 2000 th		

See question 4

4. a) Electrons are arranged in levels called _____ _____

   This arrangement is called the E_____ C_____

   b) How many electrons can ‘fit’ in the first shell? _____

   c) How many electrons can ‘fit’ in the second shell? _____

   d) How many electrons can ‘fit’ in the third shell? _____

   e) We will study only up to element 20, calcium, which has 20 electrons.

   How many electrons are in calcium's first electron shell? _____

   How many electrons are in calcium's second electron shell? _____

   How many electrons are in calcium's third electron shell? _____

   How many electrons are in calcium's fourth electron shell? _____

5. Draw diagrams for the atoms of the first twenty elements (hydrogen to calcium), showing which electron shells are being used, and dots to represent the electrons in each of the shells.

6. What is the <u>atomic number of an element</u>?

   _____

7. a) What is the atomic number of sodium? _____

   b) Does any other element have the same atomic number as sodium? _____

   c) How many elements have an atomic number of 13? _____

   d) How many elements have an atomic number of 87? _____

   e) Examine your periodic table.

   Are the elements placed in order of atomic number? _____

8. What is the mass number of an atom?

_____

_____

9. All atoms are neutral – they have no overall (net) electric charge.
Why are they neutral, even though they contain positively and negatively charged particles?

_____

_____

10. Complete the following table. You will need to refer to your periodic table and list of names and symbols of the elements. The first has been done for you.

Number of protons in atom	Number of neutrons in atom	Atomic number	Mass number	Number of electrons	Name of element	Symbol of element
12	14	12	26	12	Magnesium	$^{26}_{12}Mg$
19	24				Potassium	
7			15	7		
		9	19			
			28	13		
	16				Silicon	
20			40			
	35				Copper	
		1	1		Hydrogen	
						$^{18}_{8}O$
						$^{197}_{79}Au$